Codex Prompt: Revise Multicycle Model to One Likelihood Contribution per Patient

Model-only implementation change; no trial-design changes and no testing.

Modify only the newly added `multicycle_additive` toxicity and efficacy models so that **each patient contributes exactly one likelihood term**, corresponding to that patient's latest/current treatment cycle.

Do not modify any trial-design logic, allocation rules, recycling rules, stopping rules, futility rules, OBD selection, cohort formation, event scheduling, no-skipping logic, or other existing model options.

Do not add tests.

The existing arbitrary-cycle state recursion should be retained. The required change is that earlier cycles are used only to construct the patient's cumulative state and **must not separately contribute likelihood terms**.

1. Intended model

For patient i, suppose the patient has received C_i cycles with actual doses

$$d_{i1}, d_{i2}, \ldots, d_{iC_i}.$$

Define the uncapped toxicity state recursively by

$$S_{T,i1} = p_{T,d_{i1}},$$

and

$$\boxed{S_{T,ic} = p_{T,d_{ic}} + \alpha_T S_{T,i,c-1}, \qquad c \geq 2.}$$

Therefore,

$$S_{T,iC_i} = p_{T,d_{iC_i}} + \alpha_T p_{T,d_{i,C_i-1}} + \alpha_T^2 p_{T,d_{i,C_i-2}} + \cdots + \alpha_T^{C_i-1} p_{T,d_{i1}}.$$

The patient's toxicity probability is based only on the latest state:

$$\boxed{q_{T,i} = \min(1, S_{T,iC_i}).}$$

Similarly for efficacy,

$$S_{E,i1} = p_{E,d_{i1}},$$

$$\boxed{S_{E,ic} = p_{E,d_{ic}} + \alpha_E S_{E,i,c-1},}$$

and

$$\boxed{q_{E,i} = \min(1, S_{E,iC_i}).}$$

The historical state remains **uncapped**. Only the final/current probability is capped.

Correct:

```
S_current <- p_current + alpha * S_previous
q_current <- min(1, S_current)
```

Incorrect:

```
q_current <- min(1, p_current + alpha * q_previous)
```

2. Critical likelihood correction

The current implementation incorrectly gives every administration its own likelihood contribution. For a patient with three cycles, the current implementation effectively gives

$$L_i = L_{i1} L_{i2} L_{i3}.$$

This must be changed.

The intended likelihood is

$$\boxed{L_i = L_{i,C_i}.}$$

Therefore,

$$C_i = 1 \Rightarrow L_i = L_{i1}, \qquad C_i = 2 \Rightarrow L_i = L_{i2}, \qquad C_i = 3 \Rightarrow L_i = L_{i3}.$$

Earlier administrations are retained only to compute S_{i,C_i}. They do not have separate Bernoulli likelihood terms.

3. Required data representation

Continue storing one row per administration because the full treatment history is required to construct the state.

For each administration retain identifiers such as:

```
admin_id
patient_id
cycle
dose
previous_admin_id
previous_state_index
```

Add or construct a mapping that identifies the **latest/current administration row for each patient** at the time the model is fitted, for example:

```
latest_admin_index
```

or an equivalent structure.

If the administration history is:

```
row 1: patient A, cycle 1
row 2: patient A, cycle 2
row 3: patient A, cycle 3
row 4: patient B, cycle 1
row 5: patient B, cycle 2
row 6: patient C, cycle 1
```

then the likelihood rows are:

```
A -> row 3
B -> row 5
C -> row 6
```

while rows 1, 2, and 4 are still needed for state construction.

The model therefore needs two counts conceptually:

```
N_admin
N_patient
```

where `N_admin` is the number of administration-history rows and `N_patient` is the number of likelihood contributions.

4. Toxicity JAGS structure

Retain the CRM definition of the regular dose probabilities and the shared carryover parameter. Conceptually:

```
beta ~ dnorm(0, tau_beta)

g_alpha_1 ~ dgamma(a_r, 1)
g_alpha_2 ~ dgamma(b_r, 1)

alpha <- g_alpha_1 / (g_alpha_1 + g_alpha_2)

for (j in 1:J) {
    p[j] <- pow(q[j], exp(beta))
}
```

Construct the state for **all administration rows**:

```
state_tox[1] <- 0

for (r in 1:N_admin) {
    state_tox[r + 1] <-
        p[dose[r]] +
        alpha * state_tox[previous_state_index[r]]

    theta_admin[r] <- min(1, state_tox[r + 1])
}
```

Then define the likelihood only over patients:

```
for (i in 1:N_patient) {
    theta_patient[i] <- theta_admin[latest_admin_index[i]]
    y_patient[i] ~ dbern(theta_patient[i])
}
```

Do **not** use:

```
for (r in 1:N_admin) {
    y[r] ~ dbern(theta_admin[r])
```

```
}
```

because that gives multiple likelihood contributions to patients with multiple cycles.

5. Complete-outcome toxicity likelihood

For patient i,

$$Y_{T,i} \sim \text{Bernoulli}(q_{T,i}),$$

where

$$q_{T,i} = \min(1, S_{T,iC_i}).$$

Thus

$$\boxed{L_{T,i} = q_{T,i}^{Y_{T,i}}(1 - q_{T,i})^{1-Y_{T,i}}.}$$

The full toxicity likelihood is

$$\boxed{L_T = \prod_{i=1}^{N_{\text{patient}}} L_{T,i}.}$$

There must be exactly one toxicity likelihood contribution per patient.

6. TITE toxicity model

Apply the same one-contribution-per-patient principle to the TITE toxicity model.

All administration rows are still required to recursively construct the latest toxicity state. However, only the latest administration of each patient enters the TITE likelihood.

For patient i, let

$$q_{T,i} = \min(1, S_{T,iC_i})$$

and let $w_{T,i}$ be the TITE weight corresponding to that patient's latest/current administration.

Use

$$\boxed{L_{T,i} = (w_{T,i}q_{T,i})^{Y_{T,i}}(1 - w_{T,i}q_{T,i})^{1-Y_{T,i}}.}$$

Therefore:

- observed DLT: $q_{T,i}$, with $w_{T,i} = 1$;
- completed non-DLT: $1 - q_{T,i}$;
- pending non-DLT: $1 - w_{T,i}q_{T,i}$.

Earlier cycles must not contribute additional TITE likelihood terms.

The R data preparation should therefore pass patient-level vectors corresponding to the current/latest administration, for example:

```
y_patient
w_patient
latest_admin_index
```

while preserving the full administration-level history for recursive state construction.

7. Efficacy JAGS structure

Apply the same architecture to efficacy.

Construct all historical states:

```
state_eff[1] <- 0

for (r in 1:N_admin) {
    state_eff[r + 1] <-
        p_regular[dose[r]] +
        alpha * state_eff[previous_state_index[r]]

    theta_eff_admin[r] <- min(1, state_eff[r + 1])
}
```

Then only use the current/latest row for each patient in the likelihood. For complete binary outcomes:

```
for (i in 1:N_patient) {
    theta_eff_patient[i] <-
        theta_eff_admin[latest_admin_index[i]]

    eff_patient[i] ~ dbern(theta_eff_patient[i])
}
```

Again, do not create efficacy likelihood terms for historical cycles.

8. Delayed-outcome efficacy likelihood

For TITE/delayed efficacy, each patient contributes exactly one likelihood term based on the latest/current administration.

For patient i, define

$$q_{E,i} = \min(1, S_{E,iC_i}).$$

Let

$$\Delta_{E,i} = 1$$

if the efficacy outcome of the current/latest cycle is ascertained, either because a response has been observed or the full efficacy assessment window has completed without response. Otherwise, $\Delta_{E,i} = 0$.

Use

$$\boxed{L_{E,i} = q_{E,i}^{\Delta_{E,i} Y_{E,i}} (1 - q_{E,i})^{\Delta_{E,i}(1-Y_{E,i}) + (1-\Delta_{E,i}) w_{E,i}}}.$$

Thus the latest cycle contributes:

- observed response: $q_{E,i}$;
- completed nonresponse: $1 - q_{E,i}$;
- pending nonresponse: $(1 - q_{E,i})^{w_{E,i}}$.

If the current implementation uses the zeros trick, preserve that implementation, but apply the zeros-trick likelihood only to the `N_patient` latest/current observations rather than all `N_admin` administration rows.

9. Behavior across interim analyses

The identity of the patient's likelihood contribution changes when that patient receives another cycle.

After patient A has only cycle 1:

$$L_A = L_{A1}.$$

After A receives cycle 2:

$$L_A = L_{A2},$$

where

$$q_{A2} = \min\{1, p_{d_{A2}} + \alpha p_{d_{A1}}\}.$$

The likelihood must **not** become

$$L_{A1} L_{A2}.$$

After cycle 3:

$$L_A = L_{A3},$$

with

$$q_{A3} = \min\{1, p_{d_{A3}} + \alpha p_{d_{A2}} + \alpha^2 p_{d_{A1}}\}.$$

At each model refit, reconstruct the patient's full current history and include only the terminal/current likelihood contribution.

10. Patient-specific next-cycle prediction

Retain the current state-based prediction mechanism.

If patient i's latest uncapped state is S_{T,iC_i}, then a proposed next toxicity administration at dose k has

$$S_{T,i,C_i+1}(k) = p_{T,k} + \alpha_T S_{T,iC_i},$$

and

$$q_{T,i,C_i+1}(k) = \min\{1, S_{T,i,C_i+1}(k)\}.$$

Similarly,

$$S_{E,i,C_i+1}(k) = p_{E,k} + \alpha_E S_{E,iC_i},$$

$$q_{E,i,C_i+1}(k) = \min\{1, S_{E,i,C_i+1}(k)\}.$$

Do not change the existing recycling decision rules that use these predictions.

11. Scope restriction

This task is strictly a **model implementation correction**.

Only modify:

- `multicycle_additive` toxicity JAGS models;
- `multicycle_additive` efficacy JAGS models;
- TITE versions of those models;

- model-specific R data preparation needed to distinguish administration-history rows from patient-level likelihood rows;
- model-specific posterior prediction code if necessary to remain compatible with the new data representation.

Do **not** modify the mathematical behavior of any existing non-multicycle model.

Do **not** modify trial-design code.

Do **not** modify allocation, cohort, stopping, futility, MTD, OBD, no-skipping, enrollment, or recycling rules.

Do **not** add tests.

Full administration history determines the state, but each patient contributes only the likelihood of the latest cycle.